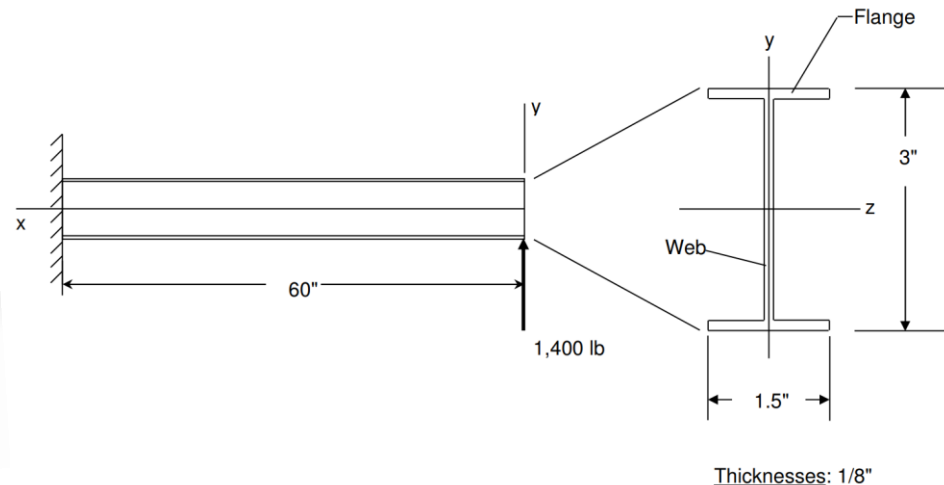
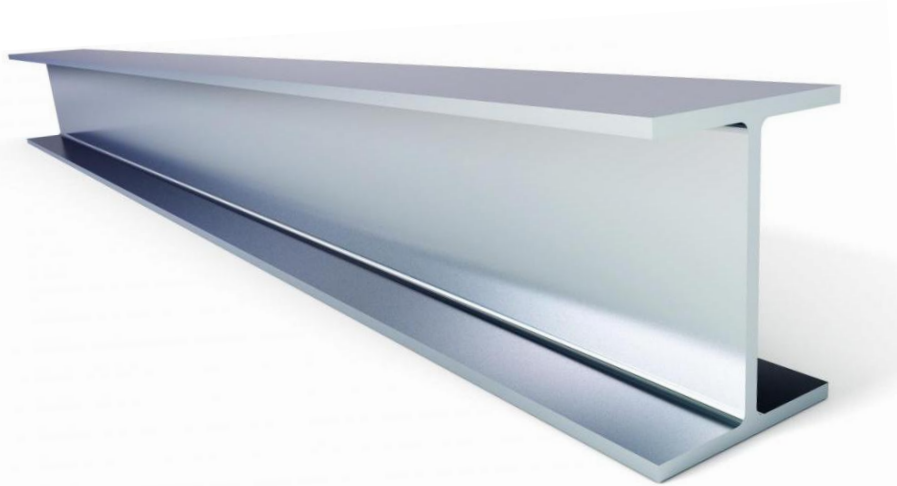


4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

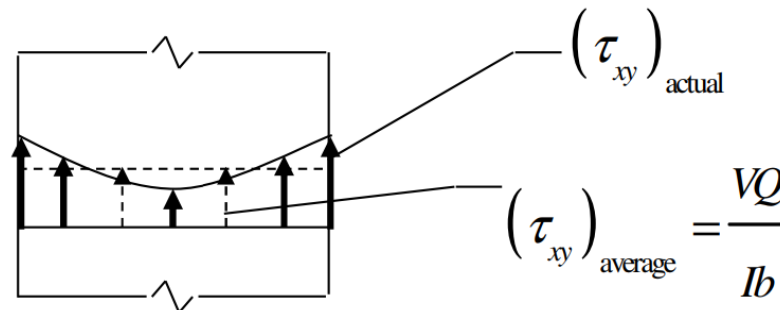
- Let's focus now on shear stress; consider the cantilevered I-beam with loading as shown:



- The formula we are familiar with is $\tau = \frac{VQ}{Ib}$ where:
 - τ = shear stress at a specific point in a beam's cross-section
 - V is the shear force on the entire cross-section
 - Q is the first moment of area of the section above the point of interest
 - I is the moment of inertia of the entire cross-section
 - b is the thickness of the cross-section at the point of interest

4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- This formula turns out to be quadratic (i.e. parabolic) in y , with τ being 0 at the top and bottom of the cross-section and a maximum at the midpoint (i.e. the neutral axis); **we will see this developed in the next few slides**
- This formula assumes τ is constant across the thickness, i.e. at a given y distance above or below the neutral axis, τ is constant at every z location
- In actuality, for the web (i.e. connecting joint) of an I-beam, τ is not constant over the thickness, but rather it varies parabolically in z as shown below:



4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- Notice that the largest value of τ occurs at the outer edges of the thickness
- For design purposes, since we are always wanting to find $\tau_{\max}$, this largest τ is of interest
- The equation $\tau = \frac{VQ}{Ib}$ does not give largest τ , but average τ across the thickness
- So does $\tau = \frac{VQ}{Ib}$ give a value close enough to largest τ for design purposes?

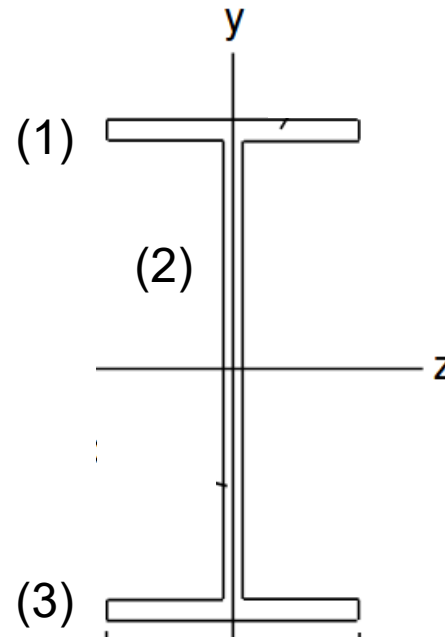


4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- **It can be shown that average τ will in fact be close to largest τ if the thickness b is small**
 - Largest τ will not exceed average τ by more than 0.8% as long as $h > 4b$
- **Consider the web of the I-beam under consideration**
 - $h = 2.75\text{in}$ and $b = (1/8)\text{in} \rightarrow h = 22b$
- **Therefore in this example we can in fact assume that τ is constant across the thickness, and we expect that $\frac{VQ}{Ib}$ will yield a value of τ_{average} very close to τ_{largest}**
- **BTW...the τ we are discussing is actually τ_{xy} , i.e. shear stress on the “x” face in the “y” direction**

4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- Let us now use $\tau_{xy} = \frac{VQ}{Ib}$ to find τ_{xy} as a function of y in both the web and the flanges of the I-beam
- For the web, $V = 1400lb$ (we could formally show this with a shear force diagram, but it's pretty easy to see!) and $b = (1/8)in$
- I is the moment of inertia of the entire cross section about the neutral axis (I_z)
- To calculate this, let us parse the cross section into three segments labeled 1-3



4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

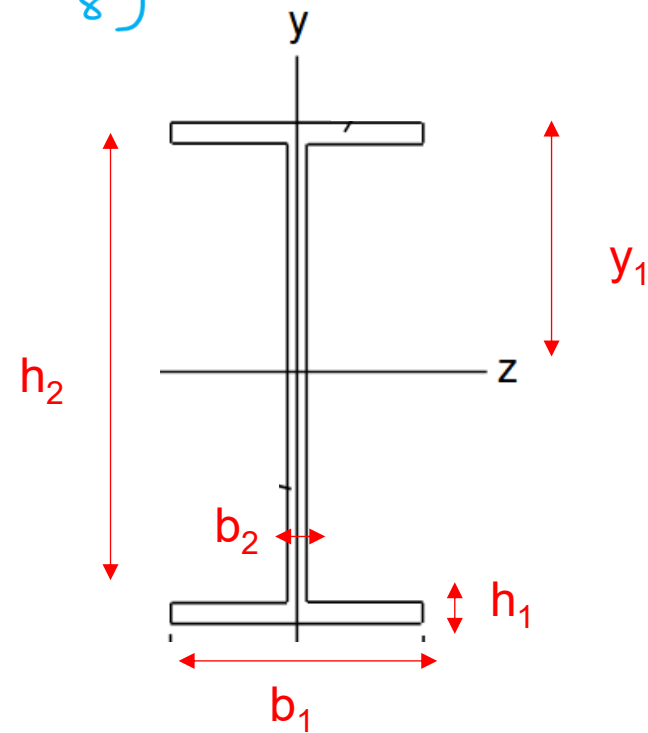
- Then our calculations for I are:

$$I = I_1 + I_2 + I_3 \xrightarrow{I_1 = I_3} I = I_2 + 2I_1$$

$$I_2 = \frac{b_2 h_2^3}{12} + A Y_2^2 = \left[\frac{1}{12} \times \frac{1}{8} \times \left(3 - 2\left(\frac{1}{8}\right) \right)^3 \right] + 0$$

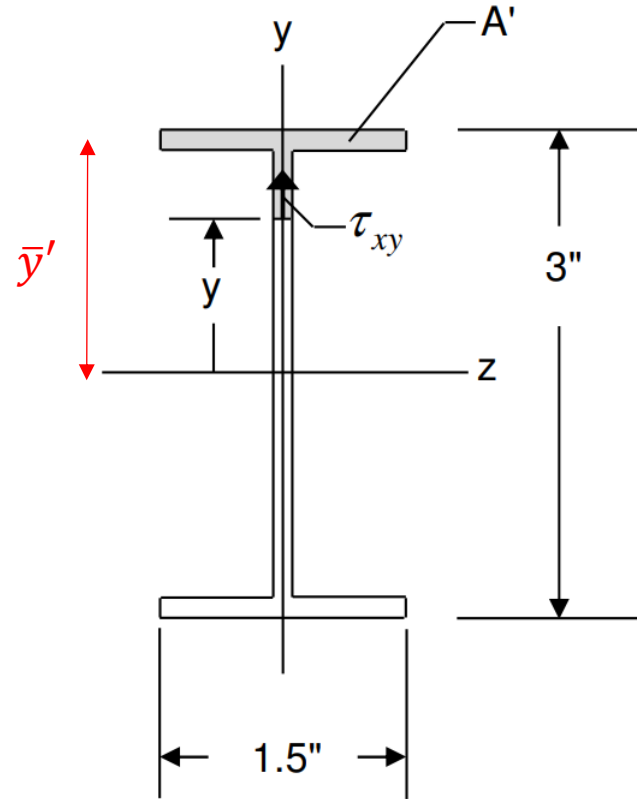
$$I_1 = \frac{b_1 h_1^3}{12} + A Y_1^2 = \left[\frac{1}{12} \times 1.5 \times \left(\frac{1}{8} \right)^3 \right] + \left[\left(1.5 \times \frac{1}{8} \right) \times \left(1.5 - 2\left(\frac{1}{8}\right) \right) \right]$$

$$I = 0.9920 \text{ in}^4$$



4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

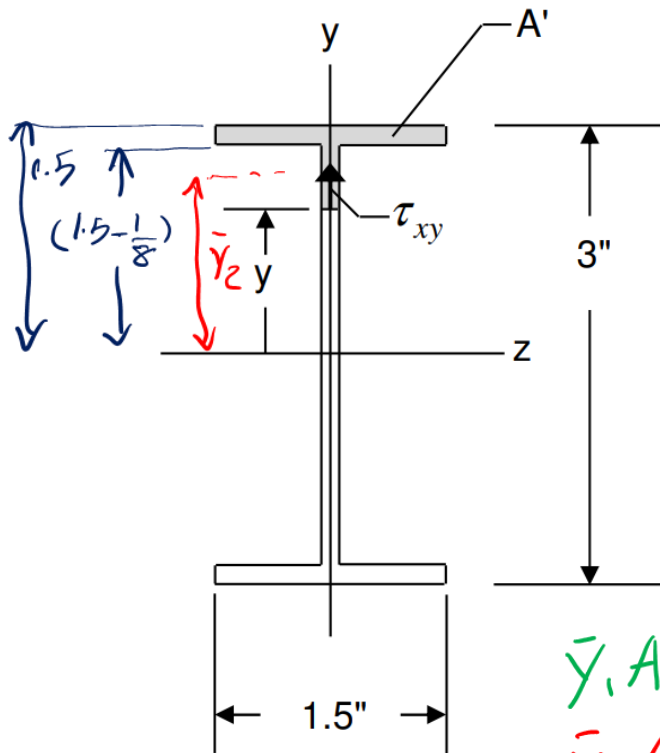
- We've now calculated V , I , and b , all of which are constant for every point in the web
- It remains to calculate Q , which will vary as a function of y
- Q is the first moment of area of the section above the point of interest
- So for a given y location along the web, $Q = \bar{y}'A'$ where A' is the area of the section above that location and $\bar{y}'$ is the y coordinate of the centroid of that section



Thicknesses: 1/8"

4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- Then our calculations for Q are:



$$Q = \bar{y}' A'$$

$$\bar{y}' = \frac{\bar{y}_1 A_1 + \bar{y}_2 A_2}{A'}$$

$$Q = \left(\frac{\bar{y}_1 A_1 + \bar{y}_2 A_2}{A'} \right) A' = \bar{y}_1 A_1 + \bar{y}_2 A_2$$

$$\bar{y}_1 A_1 = \left(1.5 - \left(\frac{1}{2} \times \frac{1}{8} \right) \right) \times \left(1.5 \times \frac{1}{8} \right)$$

$$\bar{y}_2 A_2 = \left[\frac{\left(1.5 - \frac{1}{8} \right) + y}{2} \right] \times \left[\left(1.5 - \frac{1}{8} - y \right) \times \frac{1}{8} \right]$$

Thicknesses: 1/8"



4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- Inserting all of this into $\tau_{xy} = \frac{VQ}{Ib}$ yields:

$$\Rightarrow \tau_{xy} = \frac{[1,400 \text{ lb}] \left\{ \left[\frac{\left(1.5'' - \frac{1}{8}''\right) + y}{2} \right] \left[\left(1.5'' - \frac{1}{8}''\right) - y \right] \left[\frac{1}{8}'' \right] + \left[1.5'' - \frac{1}{16}'' \right] [1.5''] \left[\frac{1}{8}'' \right] \right\}}{\underbrace{\left\{ \frac{(1.5'')(3'')^3}{12} - 2 \left[\frac{(0.6875'')(2.75'')^3}{12} \right] \right\}}_{0.9920}} \left[\frac{1}{8}'' \right]$$

$$(\tau_{xy})_{\text{Web}} = (705.6) \left\{ \left[(1.375)^2 - y^2 \right] + 4.3125 \right\}$$

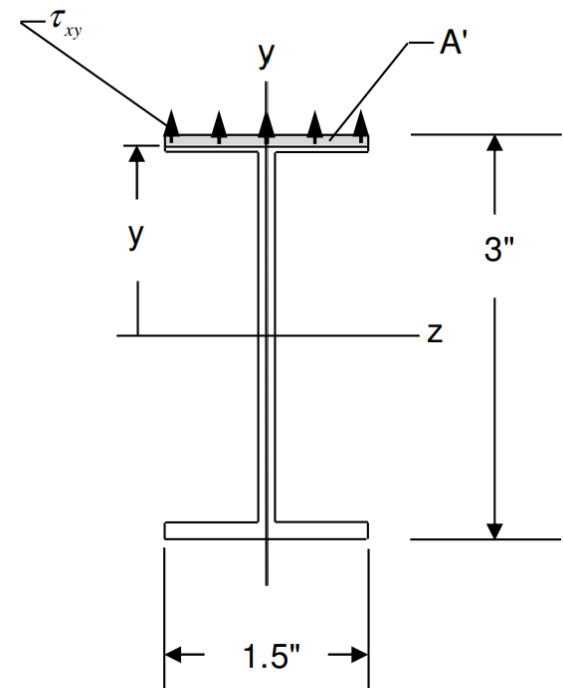
- Max value of $\tau_{xy,\text{web}}$ is 4377 psi at the midpoint of the segment ($y = 0$) and min value is 3043 psi at the top and bottom of the segment ($y = \pm 1.375\text{in}$)

4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- Let us now evaluate $\tau_{xy} = \frac{VQ}{Ib}$ for each flange of the cross section
- V and I are the same values as before, $b = 1.5\text{in}$, and $Q = \bar{y}'A'$ where y varies from $(1.5 - 1/8)\text{in}$ to 1.5in
- Inserting into $\tau_{xy} = \frac{VQ}{Ib}$ yields:

$$\tau_{xy} = \frac{(1,400 \text{ lb}) \left[\left(\frac{1.5'' + y}{2} \right) (1.5'' - y) (1.5'') \right]}{(0.9920 \text{ in}^4) (1.5'')}$$

$$(\tau_{xy})_{\text{Flanges}} = (705.6) \left[(1.5)^2 - y^2 \right]$$



Thicknesses: 1/8"

4.3: Shear Stresses in Thin-Walled Beams Due to Transverse Loads

- Therefore $\tau_{xy, \text{flange}}$ is 0 at the “outer” edge of each flange ($y = 1.5 \text{ in}$) and increases to 253.6 psi at the “inner” edge of each flange ($y = \pm 1.375 \text{ in}$)
- Now that we know the distribution of $\tau_{xy, \text{web}}$ and $\tau_{xy, \text{flange}}$, we can plot τ_{xy} as a function of y as follows:

